Probability and Statistics

The language you need before you can talk about data — Chapter 3

Amir Farbin

Lecture 6 — Probability

Why we're doing this

- We're about to do data science. What *is* data?
- To answer that we need the language of **probability and statistics**
- Starting from no assumptions — even if you've seen it, a second pass goes deeper
- Goal: look at a column of numbers and say something defensible about it

Some terms

- **Probability** — framework for quantifying the likelihood of future events
  - *(e.g. 50% chance of rain — 50% of the **area**, not of you)*
- **Statistics** — analysis of data using the language of probability
  - *(e.g. the almanac's average temperature — ten years in one number)*
- **Data** — observations / measurements of past events
- **Experiment** — controlled setup to test a specific hypothesis

Models

- **Theoretical** — captures assumptions / best understanding; little data input
- **Empirical** — no mechanism, just fits previous data and predicts *(e.g. ball-and-stick molecules)*
- **Statistical** — predicts probability of future events from previous data
- **Generative** — creates new data
  - **Simulation** — theory $\rightarrow$ new data, can be fully deterministic *(e.g. billiards)*
  - **Monte Carlo** — samples known probability distributions *(e.g. rolling dice)*

Generative AI $\approx$ an empirical Monte Carlo — learn the distributions, then sample them

Science

Propose a model $\rightarrow$ devise an experiment (aim at the **weak** spot) $\rightarrow$ predict $\rightarrow$ measure $\rightarrow$ test agreement $\rightarrow$ update $\rightarrow$ repeat

- Measurement comes back 5 ± 1, theory said 4.5 — now you need a **statistical test**
- Most of science is **eliminating bad hypotheses**
- Two hundred papers a year, one particle a decade — the value is the phase space ruled out

Frequentist and Bayesian

- Disease frequency 1 in 1000 $\rightarrow$ pick someone at random, 1 in 1000 is sick
  - **Frequentist**: probability is a long-run frequency
- You took *one* test and it's positive. You can't repeat it.
  - **Bayesian**: probability is a degree of belief

Same equations. Same computations. Same answers. Only the interpretation differs.

What an analysis actually does

Three jobs, in order:

1. **Estimate** — measure the parameter
2. **Quantify the uncertainty** on that estimate
3. **Test** the extent to which a prediction agrees with the data

What an analysis actually does

Three jobs, in order:

1. **Estimate** — measure the parameter
2. **Quantify the uncertainty** on that estimate
3. **Test** the extent to which a prediction agrees with the data

Step 2 is the one people skip. It is what makes 1 and 3 mean anything.

(later today: 13% vs a friend's 15% is only a difference once you attach $\pm 5\%$*)*

Where uncertainty comes from

- **Stochastic** — genuine underlying randomness
  - quantum mechanics is the pure case; a shuffled deck is the everyday one
- **Incomplete observations** — you can't see everything
  - Monty Hall: the answer depends entirely on what the host knows and you don't
- **Unknowns** — incomplete modeling. You don't know what you left out

...and the one you can do something about

Drop a ball, time the fall, repeat. The measurements scatter. **Why?**

- Not quantum mechanics. It's the **apparatus**
- How precisely is the drop one metre? How good is your thumb on the stopwatch?
- Start and stop with a **laser** instead $\rightarrow$ the distribution **narrows**

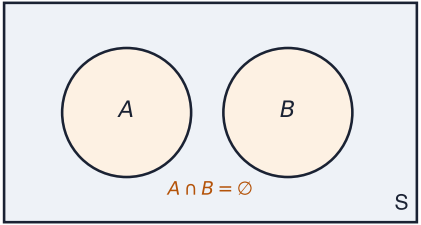
…and the one you can do something about

Drop a ball, time the fall, repeat. The measurements scatter. **Why?**

- Not quantum mechanics. It's the **apparatus**
- How precisely is the drop one metre? How good is your thumb on the stopwatch?
- Start and stop with a **laser** instead $\rightarrow$ the distribution **narrows**

You can't remove it, but you can control it — and repeating the experiment *measures* it.

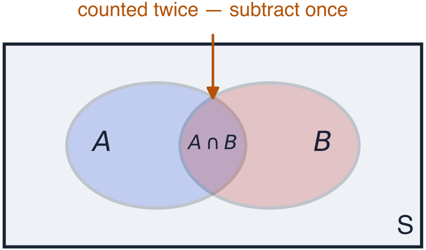
Basic definitions



Picture: drop a coin on the table. S *is the table; probability* $\propto$ *area.*

- $P(S) = 1$ — something must happen
- $0 \leq P(A) \leq 1$, $P(\bar{A}) = 1 - P(A)$, $P(\emptyset) = 0$
- $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$

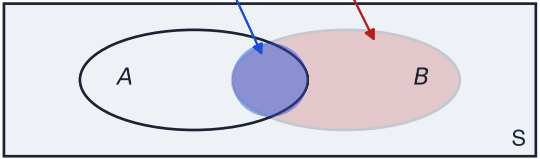
When they overlap



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$



conditioning on B throws away everything outside B,
then asks what fraction of what is left is also A

Example — the die

Rolled a six-sided die. I tell you: **three or less**. What's the chance it was a 1?

$$P(1 \mid \leq 3) = \frac{P(1 \text{ and } \leq 3)}{P(\leq 3)} = \frac{1/6}{3/6} = \frac{1}{3}$$

- Numerator is $1/6$ — a one *is* already three or less
- Denominator is $3/6$ — three faces qualify

Independence

$$P(A, B) = P(A)P(B) \quad \Longrightarrow \quad P(A \mid B) = P(A)$$

Knowing B tells you **nothing** about A.

Independence

$$P(A, B) = P(A)P(B) \quad \Longrightarrow \quad P(A \mid B) = P(A)$$

Knowing B tells you **nothing** about A.

Not the same as disjoint. If $A \cap B = \emptyset$ then $P(A \mid B) = 0$ — learning B tells you A definitely did *not* happen.

Frequentist and Bayesian, again

Now that we have $P(A \mid B)$, both readings can be stated exactly.

- **Relative frequency** — A, B are outcomes of a repeatable experiment

$$P(A) = \lim_{n \to \infty} \frac{\text{number of times the outcome is } A}{n}$$

- **Subjective** — A, B are *hypotheses*, true/false statements. $P(A)$ is your degree of belief that A is true *(e.g. do I have COVID?)*

Which side of the bar?

One test. Two conditional probabilities. **Not the same number.**

$$P(+ \mid \mathsf{sick}) \qquad \text{vs} \qquad P(\mathsf{sick} \mid +)$$

Which side of the bar?

One test. Two conditional probabilities. **Not the same number.**

$$P(+ \mid \text{sick}) \qquad \text{vs} \qquad P(\text{sick} \mid +)$$

- $P(+ \mid \text{sick})$ — the **true positive rate**. Test many known-sick people and count. Repeatable. A property of the test *(the number on the side of the box)*. **Frequentist**
- $P(\text{sick} \mid +)$ — you have *one* result and there is one of you. "**I** am sick" is a **hypothesis**. **Bayesian**

The rule that separates them

$$P(\text{data} \mid \text{hypothesis}) \;\longrightarrow\; \text{a frequency, measurable by repetition}$$

$$P(\text{hypothesis} \mid \text{data}) \;\longrightarrow\; \text{a belief — and what you actually wanted}$$

The rule that separates them

$$P(\text{data} \mid \text{hypothesis}) \;\longrightarrow\; \text{a frequency, measurable by repetition}$$

$$P(\text{hypothesis} \mid \text{data}) \;\longrightarrow\; \text{a belief — and what you actually wanted}$$

You can **measure** the first. You **want** the second.

Bayes' theorem is the machine that converts one into the other.

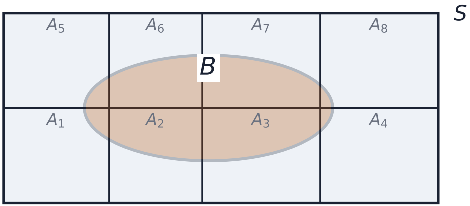
Bayes' theorem

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad P(B \mid A) = \frac{P(B \cap A)}{P(A)}$$

$$P(A \cap B) = P(B \cap A) \;\Rightarrow\; \boxed{P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}}$$

The **inverse problem**: the test gives $P(+ \mid \text{sick})$; you want $P(\text{sick} \mid +)$.

The partition



$$P(B) = \sum_i P(B \mid A_i)\, P(A_i)$$

every distinct way B could come about

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{\sum_i P(B \mid A_i)\, P(A_i)}$$

$P(A) =$ **prior** · $P(A \mid B) =$ **posterior**

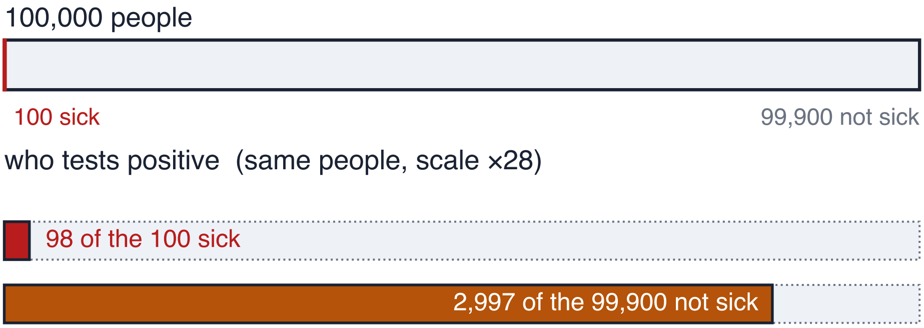
The disease test

Prior: $P(\textsf{sick}) = 0.001$, $P(\textsf{not sick}) = 0.999$

$$\begin{aligned}
P(+ \mid \textsf{sick}) &= 0.98 \quad \textsf{true positive} \\
P(- \mid \textsf{sick}) &= 0.02 \quad \textsf{false negative} \\
P(+ \mid \textsf{not sick}) &= 0.03 \quad \textsf{false positive} \\
P(- \mid \textsf{not sick}) &= 0.97 \quad \textsf{true negative}
\end{aligned}$$

$$P(\textsf{sick} \mid +) = \frac{0.98 \times 0.001}{0.98 \times 0.001 + 0.03 \times 0.999} = 0.032$$

Why 3%



the true positives are the short bar

$$P(\text{sick} \mid +) = \frac{98}{98 + 2997} = 0.032$$

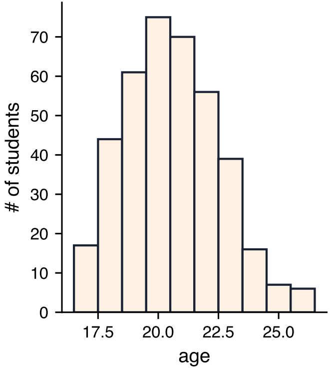
almost every positive result is a false one

Data is a table

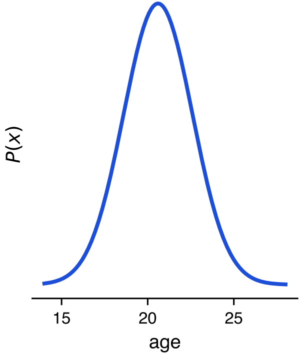
a column is a random variable

Name	Age	Major	GPA
...	...	...	...
...	...	...	...
...	...	...	...
...	...	...	...

count them into bins



the distribution underneath



- **Rows** = instances (data points) · **Columns** = features *(e.g. this class: name, age, major, GPA)*
- A column is a **random variable** — $x \sim P(x)$

Discrete and continuous

- **Discrete** $\rightarrow$ probability **mass** distribution; $P(x_1)$ is the answer directly *(e.g. a die roll, 1–6)*
- **Continuous** $\rightarrow$ $P(x = x_1) = 0$; only intervals have probability

$$P(x \in [x_1, x_1 + \delta x]) = p(x)\,\delta x$$

- Capital P = a probability · lower-case p = a **density**, needs a range
- $P(x)$ is a function; $P(x = x_1)$ is a number

Properties of any $P(x)$

- The **domain** of P must be all possible values of x
- $\forall x_i \in x : \quad 0 \leq P(x_i) \leq 1$
- **Normalisation** — it all has to add to one:

$$\sum_{x_i \in x} P(x_i) = 1 \quad \text{(discrete)} \qquad \int_{-\infty}^{\infty} p(x)\, dx = 1 \quad \text{(continuous)}$$

Both readings, now for a density

$$p(x)\,dx$$

- **Frequentist** — the *frequency* with which a measurement lands in $x \to x + dx$
- **Bayesian** — given your measurement, how likely the true x is in $x \to x + dx$

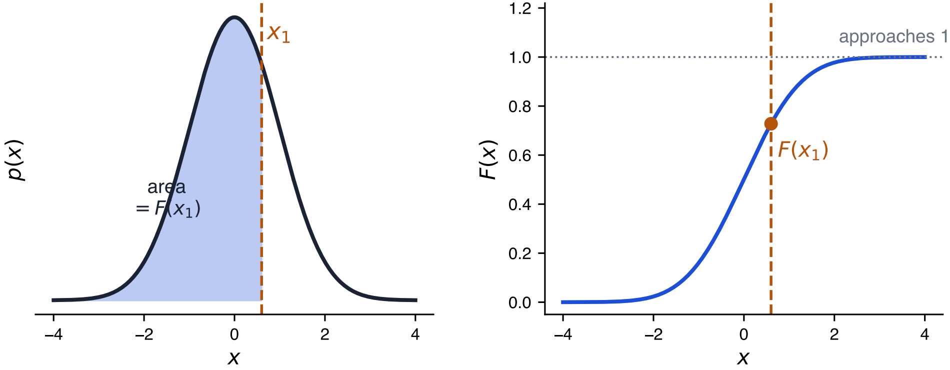
Both readings, now for a density

$$p(x)\,dx$$

- **Frequentist** — the *frequency* with which a measurement lands in $x \to x + dx$
- **Bayesian** — given your measurement, how likely the true x is in $x \to x + dx$

Same split as before, now attached to the thing you'll spend the course plotting.

Cumulative distribution



$$F(x_1) = P(x \leq x_1) = \int_{-\infty}^{x_1} p(x')\, dx'$$

Reading a percentile off it

Run it backwards: not "what fraction is below x_1?" but "**below what value is 90%?**"

$$F(x_{90}) = 0.9 \qquad \Rightarrow \qquad x_{90} = F^{-1}(0.9)$$

Reading a percentile off it

Run it backwards: not "what fraction is below x_1?" but "**below what value is 90%?**"

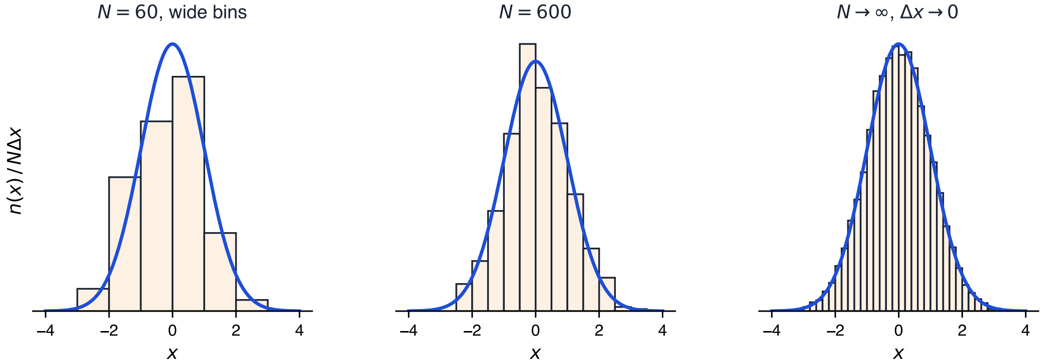
$$F(x_{90}) = 0.9 \qquad \Rightarrow \qquad x_{90} = F^{-1}(0.9)$$

From data, with no formula for F:

1. **histogram** the sample $\rightarrow$ a list of counts
2. **cumulative sum** down the list $\rightarrow$ a running total
3. find where the running total first reaches $0.9N$
4. that **bin edge** is x_{90}

Every median, quartile and percentile ever quoted is this calculation.

Histograms



$$P(x) \approx \frac{n(x)}{N\,\Delta x} \qquad P(x) = \lim_{\substack{N\to\infty \\ \Delta x\to 0}} \frac{n(x)}{N\,\Delta x}$$

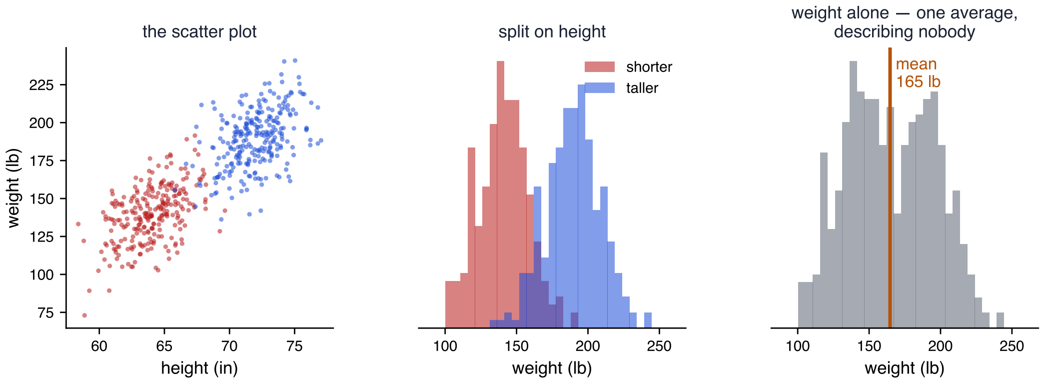
Show of hands — how many are 18? 19? That is literally how you build one.

If you know $P(x)$, you know everything

- Every prediction available from the data is in the distribution
- For one variable, a histogram is very nearly the best you can do
- Only "nearly" — if you know the functional form, fit it instead

(grab a student at random — how old are they? The histogram already answers it)

Two dimensions



First thing to do with any dataset: histogram every feature, overlaid by category — e.g. heart disease vs not. A contour plot is a topographical map.

Multivariate, formally

$$\int P(x_1, \ldots, x_n)\, dx_1 \cdots dx_n = 1 \quad \text{(normalisation)}$$

$$p(x_1) = \int P(x_1, \ldots, x_n)\, dx_2 \cdots dx_n \quad \text{(marginal)}$$

$$P(x_1, x_2) = P_1(x_1) P_2(x_2) \;\Rightarrow\; \text{independent}$$

Marginalising ***is*** *projecting: integrate over weight, get the height distribution.*

The curse of dimensionality

- Add dimensions $\rightarrow$ the space between points grows enormously
- An 18-D histogram: almost every bin empty
- An empty bin reads as probability zero. **It isn't zero — you just can't see it**
- Data needed grows exponentially
- Distances all look alike $\rightarrow$ k-**nearest neighbours** stops working; overfitting risk rises

This is why machine learning exists.

The best you can possibly do

Neyman–Pearson lemma — the most powerful test between H_0 and H_1 is the likelihood ratio

$$\Lambda(\vec{x}) = \frac{P(\vec{x} \mid H_1)}{P(\vec{x} \mid H_0)}$$

- Patient arrives, all test results are $\vec{x}$: sick or not?
- If you know the distributions, **it is impossible to do better than this**
- 1-D we can do. 2-D we can do. Beyond that we can't — *that* is data science

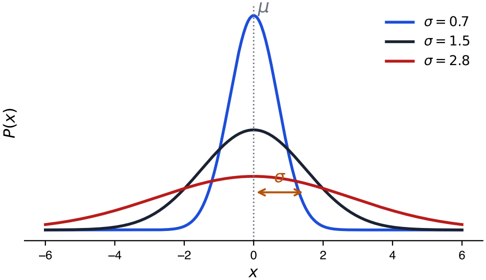
Expectation value

$$\mathbb{E}_{x\sim p}[f(x)] = \int f(x)\,P(x)\,dx$$

$$\mathbb{E}[x] = \int x\,P(x)\,dx \equiv \mu$$ — the **mean**

Variance

$$V[x] = \mathbb{E}\left[(x-\mu)^2\right] = \mathbb{E}[x^2] - \mu^2 \equiv \sigma^2$$



Covariance and correlation

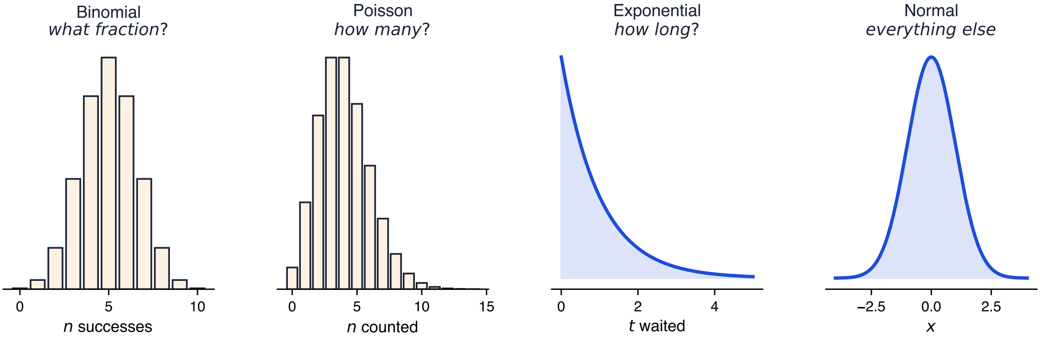
$$\mathrm{cov}[x, y] = \mathbb{E}[xy] - \mu_x \mu_y = \mathbb{E}\big[(x - \mu_x)(y - \mu_y)\big]$$

$$\rho_{xy} = \frac{\mathrm{cov}[x, y]}{\sigma_x \sigma_y}$$

$\rho = 0.5$: move x up one σ_x, and y most likely moves up half a σ_y.

Lecture 7 — Distributions, and computing with them

Distributions with names



Know which one you have $\rightarrow$ measure its parameters $\rightarrow$ predict, compare, model.

(e.g. a coin — you know the distribution before you start; you only need p)

Binomial — *what fraction?*

$$f(n \mid N, p) = \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n} \qquad \mathbb{E}[n] = Np \qquad V[n] = Np(1-p)$$

For the **rate** $\hat{p} = n/N$:

$$\boxed{\sigma_{\hat{p}} = \frac{1}{\sqrt{N}} \sqrt{p(1-p)}}$$

TPR, FPR, efficiency, conversion rate — every “what fraction of…” is binomial.

Worked: the Super Bowl

45 students in the room, 6 watched.

$$\hat{p} = \frac{6}{45} = 0.13 \qquad \sigma_{\hat{p}} = \frac{1}{\sqrt{45}}\sqrt{0.13 \times 0.87} = 0.05$$

0.13 ± 0.05

Worked: the Super Bowl

45 students in the room, 6 watched.

$$\hat{p} = \frac{6}{45} = 0.13 \qquad \sigma_{\hat{p}} = \frac{1}{\sqrt{45}}\sqrt{0.13 \times 0.87} = 0.05$$

0.13 ± 0.05

Another university reports 15% and says they like football more. **Do you believe them?**

Poisson — *how many?*

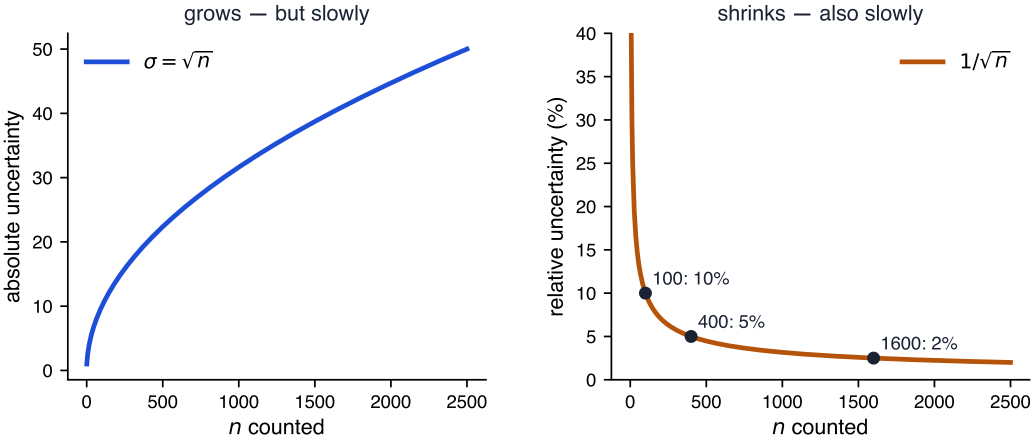
Binomial with $N \to \infty$, $p \to 0$, $Np \to \nu$:

$$f(n;\nu) = \frac{\nu^n}{n!}e^{-\nu} \qquad \mathbb{E}[n] = V[n] = \nu \qquad \sigma = \sqrt{\nu}$$

$$\boxed{n \pm \sqrt{n}}$$

(e.g. customers per hour · students through a door · radioactive decays · photons on a detector)

Counting statistics



Relative uncertainty $= 1/\sqrt{n}$ — to halve it you need **four times** the data.

Exponential — *how long?*

$$f(t;\tau) = \frac{1}{\tau} e^{-t/\tau} \quad (t \geq 0) \qquad \mathbb{E}[t] = \tau \qquad \sigma = \tau$$

(e.g. the time until the next customer arrives)

Poisson counts events in a window; exponential is the wait between them.

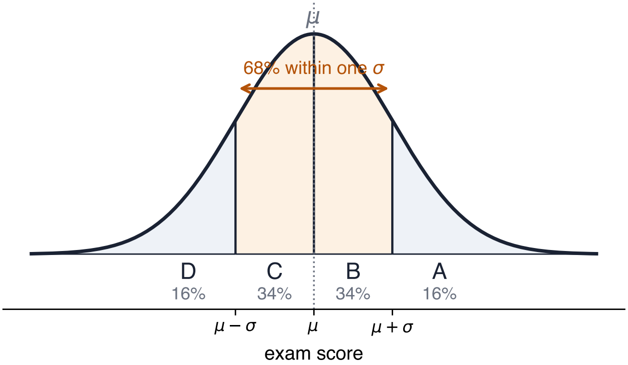
Normal — *everything else*

Central limit theorem: combine random variables $\rightarrow$ normal

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(x-\mu)^2/2\sigma^2}$$

Completely specified by μ and σ. Know those two and you know **everything**.

Curving the grades



…but look at your data

- Data often isn't normal even when it "should" be
- Usually: **several normals superimposed**
- Smokers and non-smokers. Data science and non-data science students
- The mean of the sum describes neither

...but look at your data

- Data often isn't normal even when it "should" be
- Usually: **several normals superimposed**
- Smokers and non-smokers. Data science and non-data science students
- The mean of the sum describes neither

Including the exam scores on the last slide — plot a class's marks and you often get two bumps: the students who kept up, and the students who didn't

Always plot the histogram before you quote a mean.

Computing from data

$$\mu = \int x P(x)\, dx \;\approx\; \sum_i x_i \frac{n(x_i)}{N\Delta x}\Delta x \;=\; \frac{1}{N}\sum_i x_i\, n(x_i)$$

$$\boxed{\mu = \frac{1}{N} \sum_{x_i \in \text{dataset}} x_i}$$

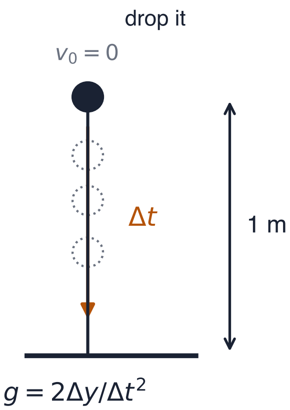
The familiar average — *because* the histogram estimates the distribution.

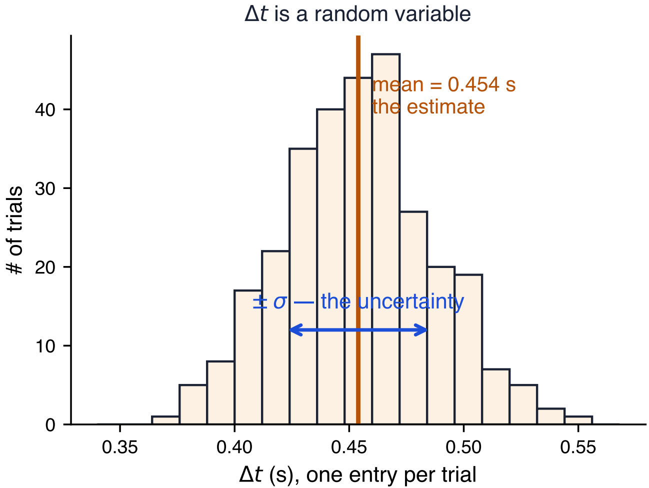
Variance, and why $N - 1$

$$\sigma = \sqrt{\frac{1}{N-1} \sum_i (x_i - \mu)^2}$$

- You needed μ to compute this — and you **estimated** μ from the same data
- That spent one **degree of freedom**
- **Bessel correction**. Matters when N is small, which is when you should be careful

A worked measurement





$$2\Delta y$$

Running it backwards

So far: **data in $\rightarrow$ what distribution did it come from?**

Running it backwards

So far: **data in → what distribution did it come from?**

Now the other direction: **given a distribution → produce data that follows it.**

- That is the **Monte Carlo simulation** from the first lecture
- Test a method before you have real data
- Ask what a result would look like *if* a hypothesis were true
- At the far end of this idea: what a generative model does

The whole problem: you have uniform random numbers between 0 and 1. How do you turn them into draws from *any* distribution?

Generating random numbers

Everything starts from one distribution — the **uniform**, flat across its range:

$$P(x) = \frac{1}{\Delta x}$$

Get that, and inversion or accept/reject turns it into anything else.

Generating random numbers

Everything starts from one distribution — the **uniform**, flat across its range:

$$P(x) = \frac{1}{\Delta x}$$

Get that, and inversion or accept/reject turns it into anything else.

- Natural entropy (noise, quantum) $\rightarrow$ blocking, rate-limited
- **Pseudo-random**: seed $\rightarrow$ deterministic sequence $\rightarrow$ repeats eventually

$$x_{n+1} = (a x_n + b) \bmod m$$

- Same seed $\rightarrow$ same sequence. That is a **feature**: reproducible simulations

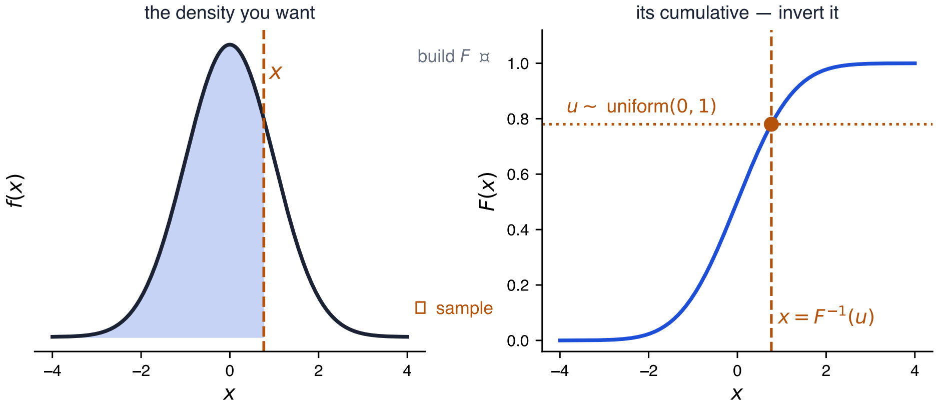
From uniform to anything — inversion

$$F(x) = \int_{-\infty}^{x} f(x')\,dx' \qquad y \sim \text{uniform} \qquad x = F^{-1}(y) \;\Rightarrow\; x \sim f(x)$$

Worked for the exponential:

$$F(T) = 1 - e^{-T/\tau} = u \qquad \Rightarrow \qquad T = -\tau \ln(1-u)$$

Why inversion works



$F(x) =$ the probability lying to the **left** of x. Pick a *share of probability* uniformly; F^{-1} says which x owns it.

Tall f $\rightarrow$ *steep* F $\rightarrow$ *a wide band of* u *lands in a narrow band of* x.

Histogramming, by hand

A histogram **is** two lists: bin edges, and a count per bin.

1. find `x_min`, `x_max` (if not given); `n_bins` defaults to 10
2. `bin_size = (x_max - x_min) / n_bins`
3. make a list of `n_bins` zeros
4. for each value, find its bin, increment, `break`

…but you usually can't invert

$x = F^{-1}(u)$ needs $F(x) = \int_{-\infty}^{x} f(x')\,dx'$ *done analytically*

…but you usually can't invert

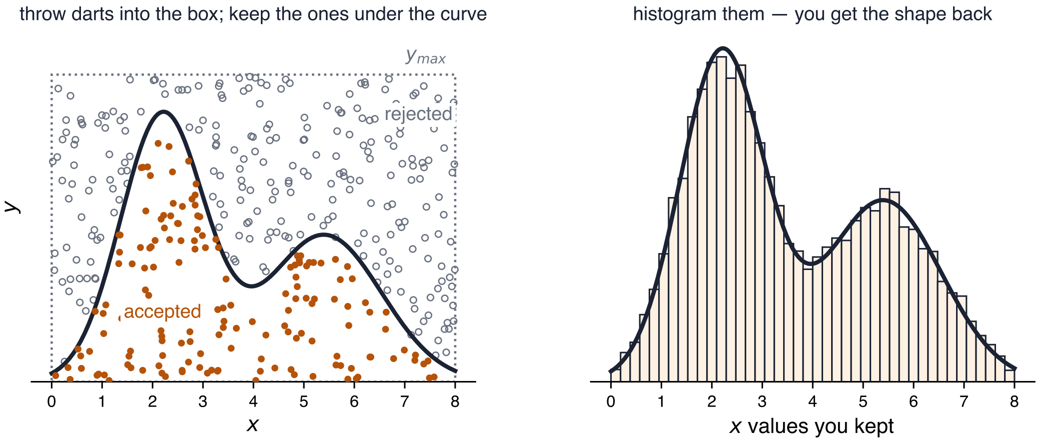
$$x = F^{-1}(u) \qquad \text{needs} \qquad F(x) = \int_{-\infty}^{x} f(x')\,dx' \quad \textit{done analytically}$$

Most functions you meet you can **evaluate** but not **integrate** in closed form.

- Inversion is off the table
- You still want to sample

Accept/reject needs no calculus at all — only the ability to evaluate f.

Accept / reject — the picture



Throw darts into the box. Keep the ones that land **under the curve**.

Why the kept values follow f

Take a thin vertical strip at some x:

- darts in it are spread **uniformly** up the box
- so the fraction surviving is the fraction of the strip under the curve
- that fraction is $f(x)/y_{max}$ — **proportional to the height of f there**

Why the kept values follow f

Take a thin vertical strip at some x:

- darts in it are spread **uniformly** up the box
- so the fraction surviving is the fraction of the strip under the curve
- that fraction is $f(x)/y_{max}$ — **proportional to the height of f there**

Tall regions keep more darts, low regions keep fewer, in exact proportion to f. That *is* sampling from f.

The trade

Inversion

- *needs:* an integral you can do analytically
- *costs:* nothing — one uniform draw per sample

Accept/reject

- *needs:* only that you can evaluate f
- *costs:* every rejected dart

The trade

Inversion

- *needs:* an integral you can do analytically
- *costs:* nothing — one uniform draw per sample

Accept/reject

- *needs:* only that you can evaluate f
- *costs:* every rejected dart

A narrow spike in a wide box $\rightarrow$ throw away 99 darts to keep 1.

Choosing between them is a real decision.

Accept / reject — the algorithm

1. range of x: $x_{min} \rightarrow x_{max}$
2. draw $x \sim$ uniform(x_{min}, x_{max})
3. range of y: $0 \rightarrow y_{max}$
4. draw $y \sim$ uniform$(0, y_{max})$
5. if $y < P(x) \rightarrow$ **accept**, else reject — until you have N

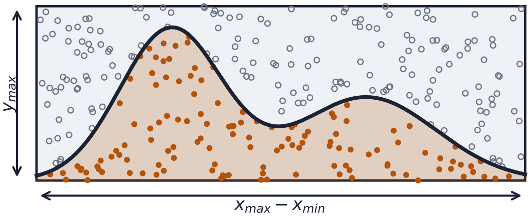
Accept / reject — the algorithm

1. range of x: $x_{min} \rightarrow x_{max}$
2. draw $x \sim$ uniform(x_{min}, x_{max})
3. range of y: $0 \rightarrow y_{max}$
4. draw $y \sim$ uniform$(0, y_{max})$
5. if $y < P(x) \rightarrow$ **accept**, else reject — until you have N

Check it worked: histogram the values you kept. You should get back the shape of $P(x)$.

…and you get integration for free

$$\text{area under } f \;=\; A \times \frac{\#\,\text{accepts}}{\#\,\text{trials}}$$



Monte Carlo integration — the same loop that samples also integrates, by counting.

Why it matters, and how good it is

- **No calculus.** Works on anything you can evaluate
- Unlike grid methods, the cost does **not explode with dimensions**
  - $\rightarrow$ the standard tool for high-dimensional integrals, i.e. most real ones

Why it matters, and how good it is

- **No calculus.** Works on anything you can evaluate
- Unlike grid methods, the cost does **not explode with dimensions**
  - $\rightarrow$ the standard tool for high-dimensional integrals, i.e. most real ones

Accuracy is just counting statistics from the first half:

$$n \text{ accepts } \Rightarrow \sqrt{n} \text{ uncertainty } \Rightarrow \frac{1}{\sqrt{n}} \text{ relative}$$

One more decimal place = a hundred times as many darts. Same law as the Super Bowl.

Summary

- Probability, both readings — and Bayes to move between them
- A histogram **estimates** a distribution: $P(x) \approx n(x)/N\Delta x$
- Know the distribution and you know everything — Neyman–Pearson makes that precise
- Beyond 2-D you can't measure it. That is the whole problem
- Mean, variance, and the error bar on every fraction you quote
- **Always look at your data**